

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

*I Ahamed kaif S (192321128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
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Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = 150 MB = $150 \times 10^6 = 150,000,000$ bytes
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: Number of segments

$$150000000 / 1500 = 100000 \text{ segments}$$

Since one segment becomes one frame,

$$\text{Frames} = 100,000$$

Step 2: Total overhead

Overhead per frame = 40 bytes

$$100000 \times 40 = 4,000,000 \text{ bytes}$$

$$\text{Total overhead} = 4 \text{ MB}$$

Step 3: Total transmitted data

$$150000000 + 4000000 = 154000000 \text{ bytes}$$

Convert to bits

$$154000000 \times 8 = 1,232,000,000 \text{ bits}$$

Step 4: Transmission time

Bandwidth

$$\begin{aligned} 50 \text{ Mbps} &= 50,000,000 \text{ bps} \\ \frac{1,232,000,000}{50,000,000} &= 24.64 \text{ s} \end{aligned}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Total overhead = **4,000,000 bytes (4 MB)**
- Transmission time = **24.64 seconds**

Explanation

- **Transport Layer** divides data into segments, provides sequencing, acknowledgements, and retransmission for reliable delivery.
- **Data Link Layer** encapsulates segments into frames, performs error detection using CRC, and retransmits corrupted frames.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = 6
- Total frames = 48

Number of windows

$$48 \div 6 = 8$$

Retransmissions

One frame lost per window

$$8 \times 1 = 8$$

Answer

- Windows required = **8**
- Retransmissions = **8 frames**

Explanation

Flow control prevents the sender from transmitting more frames than the receiver can process, reducing congestion and improving throughput.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- Packet size = 12,000 bytes
- MTU = 1500 bytes
- Header = 20 bytes

Payload per fragment

$$1500 - 20 = 1480 \text{ bytes} \quad 1500 - 20 = 1480 \text{ bytes}$$

Number of fragments

$$12000 / 1480 = 8.11 \quad 12000 / 1480 = 8.11$$

Round up

= 9 fragments

Header overhead

$$9 \times 20 = 180 \text{ bytes} \quad 9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Explanation

The Network Layer fragments packets when MTU is smaller and uses fragment offset, identification, and MF flag so the destination can correctly reassemble them.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer

- Windows = **8**
- Retransmissions = **8 frames**

Explanation

Flow control matches sender speed with receiver capacity, avoiding overflow and packet loss.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- Video = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints before failure

$$390 \div 60 = 6.5 \quad 390 \div 60 = 6.5$$

Completed checkpoints

= **6**

Checkpoint locations

60,120,180,240,300,360 MB

Retransmission

Failure at 390 MB

Resume from 360 MB

Retransmit

$$390 - 360 = 30 \text{ MB} \quad 390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation

Synchronization checkpoints allow communication to resume from the last saved point instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

Original = 900 MB

Compression = 40%

Encryption overhead = 5%

Compressed size

$$900 \times 0.60 = 540 \text{ MB} \quad 900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB} \quad 540 \times 1.05 = 567 \text{ MB}$$

Reduction

$$900 - 567 \quad 900 \times 100 = 37\% \quad 900 - 567 \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**

- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation

The Presentation Layer compresses data to reduce bandwidth usage and encrypts data to ensure confidentiality and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

Requests

$$3000 \div 3 = 1000 \quad 3000 \div 3 = 1000$$

Transmission time

Bits

$$3000 \times 8 = 24000 \text{ Mb} \quad 3000 \times 8 = 24000 \text{ Mb} \quad 24000 \div 120 = 200 \text{ seconds} \quad 24000 \div 120 = 200 \text{ seconds}$$

Answer

- Requests = **1000**
- Time = **200 seconds**

Explanation

The Application Layer provides services such as FTP, authentication, file management, and reliable file transfer for users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay

experienced by the packet and explain how each layer contributes to the communication process.

Processing delays

Application =4 ms

Transport =3 ms

Network =5 ms

Data Link =2 ms

Physical =1 ms

Total processing

$$4+3+5+2+1=15\text{ ms}$$

Propagation

18 ms

Transmission

12 ms

Total delay

$$15+18+12=45\text{ ms}$$

Answer

Total end-to-end delay = 45 ms

Explanation

- Application Layer prepares data.
- Transport Layer segments and ensures reliability.
- Network Layer performs routing.
- Data Link Layer frames data and detects errors.
- Physical Layer transmits bits over the medium.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and

the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

Useful data

80 MB

Frame size

1600 bytes

Overhead

80 bytes

Useful payload

$$1600 - 80 = 1520 \text{ bytes} \quad 1600 - 80 = 1520 \text{ bytes}$$

Frames

$$\frac{80000000}{1520} = 52631.58 \quad \frac{52080000000}{1520} = 52631.58$$

Round up

= **52,632 frames**

Total overhead

$$52632 \times 80 = 4,210,560 \text{ bytes} \quad 52632 \times 80 = 4,210,560 \text{ bytes}$$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\% \quad \frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**
- Overhead = **4,210,560 bytes**
- Efficiency = **95%**

Explanation

Protocol overhead carries addressing, error checking, and control information. Higher overhead decreases transmission efficiency by reducing the proportion of useful data.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**,

the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

Original

240 MB

Compression

30%

Segment size

1400 bytes

Frame overhead

60 bytes

Compressed size

$$240 \times 0.70 = 168 \text{ MB} \quad 240 \times 0.70 = 168 \text{ MB}$$

Bytes

$$168000000 \text{ bytes} \quad 168000000 \text{ bytes}$$

Segments

$$\begin{aligned} 168000000 / 1400 &= 120000 \\ 120000 \times 60 &= 7200000 \end{aligned}$$

Protocol overhead

$$120000 \times 60 = 7,200,000$$

$$\text{bytes } 120000 \times 60 = 7,200,000 \text{ bytes}$$

Final transmitted data

$$168000000 + 7200000 = 175200000$$

$$\text{bytes } 168000000 + 7200000 = 175200000 \text{ bytes}$$

Answer

- Compressed file = **168 MB**
- Segments = **120,000**
- Protocol overhead = **7,200,000 bytes (7.2 MB)**
- Final transmitted data = **175,200,000 bytes (175.2 MB)**

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data and ensures reliable delivery.
- **Network Layer:** Routes packets to the destination.
- **Data Link Layer:** Frames packets and performs error detection.
- **Physical Layer:** Transmits the final bit stream over the communication medium.

